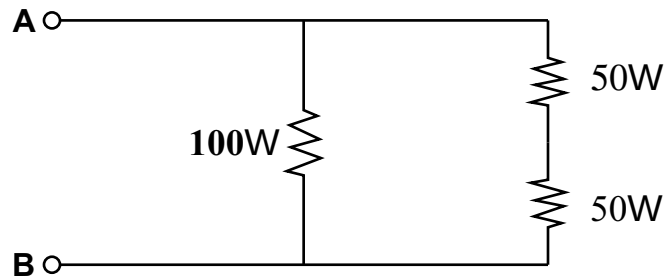


Q.22 Calculate the equivalent resistance between the terminal A and B.



SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Define inductance of an inductor. Find the series and parallel equivalent of two inductors of values L_1 and L_2 . What is the energy stored in an inductor of value 5H and the initial current flowing through it is 10 amp?

Q.24 Explain the construction, working principle and applications of Lead-Acid and Nickel-Cadmium batteries.

Q.25 Write a short note on each of the followings

- Norton theorem
- Parallel resonance circuit
- Ohm's law
- KCL

No. of Printed Pages : 4
Roll No.

221521

2nd Sem / Instrumentation & Control Engg.

Subject : Fundamentals of Electrical Engineering

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following statement is correct for an electric circuit?

- If resistance increase, voltage across it decreases.
- If resistance decreases, current through it increases.
- If the length of the conductor increases, resistance decreases.
- None of the above

Q.2 The power factor for a pure resistive circuit is

- 0
- 1
- 0.5
- 0.25

Q.3 If two resistance of 3 and 6 Ohm are in parallel, then the equivalent resistance is

- 9 Ohm
- 3 Ohm
- 6 Ohm
- 2 Ohm

- Q.4 The energy stored in an inductor is directly proportional to
- Current through it.
 - Inductance
 - Square of the inductance
 - None of the above

- Q.5 A_____ cell is a type of electrochemical cell that produces electricity through a chemical reaction that is not reversible.
- Primary
 - Secondary
 - Li ion cell
 - None of the above

- Q.6 At series resonance.
- The impedance is maximum
 - The current is minimum
 - The current is maximum
 - Both (b) and (c)

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Expand KCL.
- Q.8 KVL works on the principle of conservation of charge. (T/F)
- Q.9 Three resistance of 3 Ohm each are connected in star, the delta equivalent resistance are_____ Ohm each.

(2)

221521

- Q.10 Expand EMF.
- Q.11 Write the Ohm's law.
- Q.12 If the amplitude of an AC signal is 282 Volts then the RMS value of the AC signal is_____ volts.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 State superposition theorem. Also write the condition for a circuit to follow this theorem.
- Q.14 For an AC signal, $f(t) = 5.\sin(157t+60^\circ)$ Volts, calculate the values of
- Amplitude in volts
 - Frequency in Hz
 - Time period in seconds
 - Phase difference in degrees
- Q.15 Explain the construction and working principle of Lead-acid battery.
- Q.16 Write any four applications of solar panels.
- Q.17 Explain the principle of self and mutual inductances.
- Q.18 Define power factor and active power.
- Q.19 What are the difference between series and parallel resonance circuit?
- Q.20 What is primary and secondary cells? Also write the differences between them.
- Q.21 Explain star-delta connection.

(3)

221521